COL703/7203

12th October

i) $t_1 = t_2 \vdash t_2 = t_1$

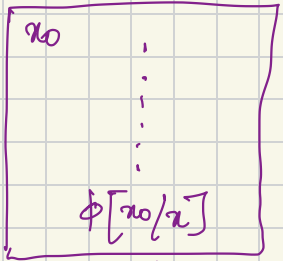
1. $t_1 = t_2$ premise
2. $t_1 = t_1$ $=i$ (axiom)
3. $(x = t_1)[t_1/x]$ same as (2)
4. $(x = t_1)[t_2/x]$ $=e$ 1, 3
5. $t_2 = t_1$ same as (4)

ii) $t_1 = t_2 , t_2 = t_3 \vdash t_1 = t_3$

1. $t_1 = t_2$ premise
2. $t_2 = t_3$ premise
3. $(t_1 = x)[t_2/x]$ same as (1)
4. $(t_1 = x)[t_3/x]$ = e , 2 , 3
5. $t_1 = t_3$ same as (4)

Proof rules for universal quantification

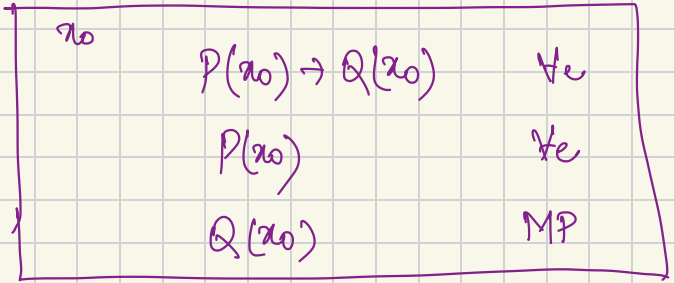
$$\frac{\forall x\, \phi}{\phi[t/x]}\ \forall e$$



$\forall i$

If x_0 is arbitrary ϕ must hold for all x's.

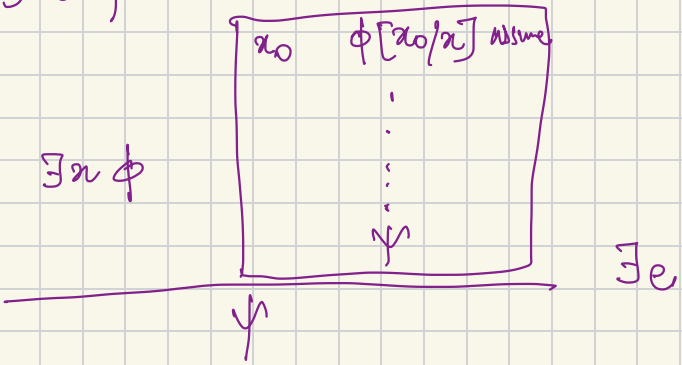
Example $\forall x\,(P(x) \to Q(x)),\ \forall x\,P(x) \vdash \forall x\,Q(x)$



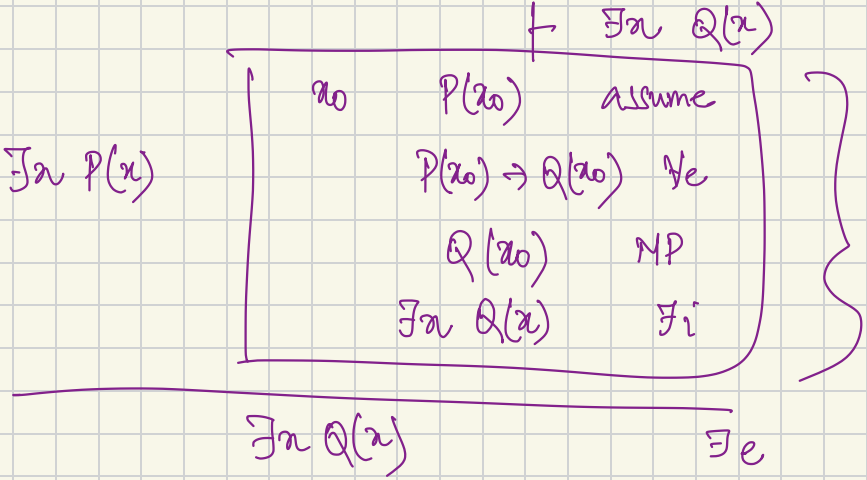
Exercise $P(t), \forall x (P(x) \Rightarrow \neg Q(x)) \vdash \neg Q(t)$

Proof rules for existential quantification

$$\frac{\phi[t/x]}{\exists x\, \phi}\ \exists i$$



Exercise $\quad \forall x\,(P(x) \to Q(x)),\ \exists x\,P(x) \vdash \exists x\,Q(x)$



$$\forall x\,(Q(x) \to R(x)),\ \exists x\,(P(x) \wedge Q(x)) \vdash \exists x\,(P(x) \wedge R(x))$$

x_0	$P(x_0) \wedge Q(x_0)$	assume
	$P(x_0)$	$\wedge e$
	$Q(x_0)$	$\wedge e$
	$Q(x_0) \to R(x_0)$	$\forall e$
	$R(x_0)$	MP
	$P(x_0) \wedge R(x_0)$	$\wedge i$
	$\exists x\,(P(x) \wedge R(x))$	$\exists i$

Quantifier Equivalences

Not all birds can fly.

$$\exists x \left(B(x) \wedge \neg F(x) \right)$$

$$\neg \forall x \left(B(x) \rightarrow F(x) \right)$$

i) $\neg \forall x \, \phi \dashv\vdash \exists x \, \neg \phi$

ii) $\neg \exists x \, \phi \dashv\vdash \forall x \, \neg \phi$

$$\neg\forall x\,\phi \;\vdash\; \exists x\,\neg\phi \qquad \text{(for any } \mathcal{M}\text{)}$$

$$\neg\forall x\,P(x) \;\vdash\; \exists x\,\neg P(x)$$

$$\neg(\phi_1 \vee \neg\phi_1) \;\vdash\; \neg\phi_1 \wedge \neg\neg\phi_1$$

$\neg(p_1 \wedge p_2)$ premise

$\neg(\neg p_1 \vee \neg p_2)$ assumption

$\neg p_1$ assumption

$\neg p_1 \vee \neg p_2$ $\vee i$

$\bot$

p_1 $\neg\neg$

$\neg p_2$ assumption

$\neg p_1 \wedge \neg p_2$ $\vee i$

$\bot$

p_2 $\neg\neg$

$p_1 \vee p_2$ $\vee i$

$\bot$

$\neg p_1 \wedge \neg p_2$ $\neg e$

$\neg \forall x P(x)$

$\neg \exists x \neg P(x)$ assumption

x_0

$\neg P(x_0)$ assume

$\exists x \neg P(x)$

$\bot$

$P(x_0)$ $\neg\neg$

$\forall x P(x)$

$\bot$

$\exists x \neg P(x)$ $\neg e$

$\neg \forall x\, P(x)$

$\neg \exists x \neg P(x)$ assumption

x_0

$\neg P(x_0)$ assume

$\exists x \neg P(x)$

$\bot$

$P(x_0)$

$\forall x P(x)$

$\bot$

$\exists x \neg P(x)$

$\neg \forall x\, \phi$

$\neg \exists x \neg \phi$ assumption

x_0

$\neg \phi[x_0/x]$ assume

$\exists x \neg \phi$

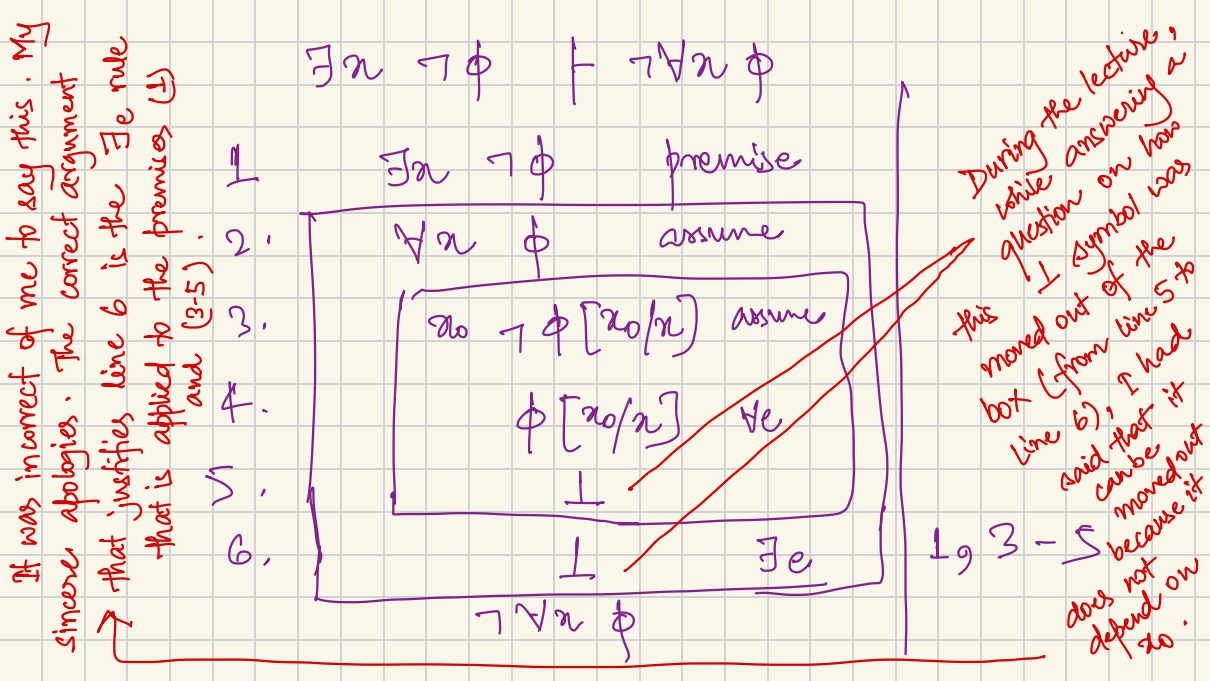
$\bot$

$\phi[x_0/x]$

$\forall x \phi$

$\bot$

$\exists x \neg \phi$


∃x ¬φ ⊢ ¬∀x φ
1. ∃x ¬φ premise
2. ∀x φ assume
3. x0 ¬φ[x0/x] assume
4. φ[x0/x] ∀e
5. ⊥
6. ⊥ ∃e
¬∀x φ
1, 3-5
It was incorrect of me to say this. My sincere apologies. The correct argument that justifies line 6 is the ∃e rule that is applied to the premises (1) and (3-5).
During the lecture, while answering a question on how this ⊥ symbol was moved out of the box (from line 5 to line 6), I had said that it can be moved out because it does not depend on x0.

Exercise $(\forall x\ \phi) \wedge \psi \vdash \forall x\,(\phi \wedge \psi)$

(assuming that x is not free in ψ)

$\forall x\,\phi \wedge \psi$ premise

x_0.

$\phi[x_0/x]$

ψ

$\psi[x_0/x]$ (same as above because x is not free in ψ)

$\phi[x_0/x] \wedge \psi[x_0/x]$

$(\phi \wedge \psi)[x_0/x]$